ALM - 2

[CO - 1]

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Date: 06/01/26

Section: 7

* Quiz/Test Questions:

1) what is the simplest measure of dispersion?

Ans. The Range is the simplest measure of dispersion. It is calculated as:

Range = Maximum value − Minimum value.

2) Explain the concept of Standard Deviation.

Ans. Standard deviation is a measure of dispersion that shows how much individual observations differ from the mean value. A low standard deviation indicates clustered data, while a high value indicates widely spread data.

3) Calculate the variance and Standard Deviation for: 2, 4, 6, 8, 10, 12, 14.

Ans. Mean $= \Sigma x/n = 56/7 = 8$

Variance $(\sigma^2) = \Sigma (x - \bar{x})^2/n = 112/7 = 16$

Standard Deviation $(\sigma) = \sqrt{\sigma^2} = \sqrt{16} = 4$

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4> Which measure is most affected by extreme values?

a) Range b) Variance c) Standard Deviation

d) All of the above

Ans. The correct answer is d) All of the above. Range, variance, and standard deviation are all affected by extreme values because they depend on maximum, minimum & squared deviations from the mean.

5> If all observations in a dataset are equal, the standard deviation is how much?

Ans. If all observations are equal, each value equals the mean, so deviations are zero. Therefore, variance becomes zero and the standard deviation is also zero, ~~indication~~ indicating no dispersion in the dataset.

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